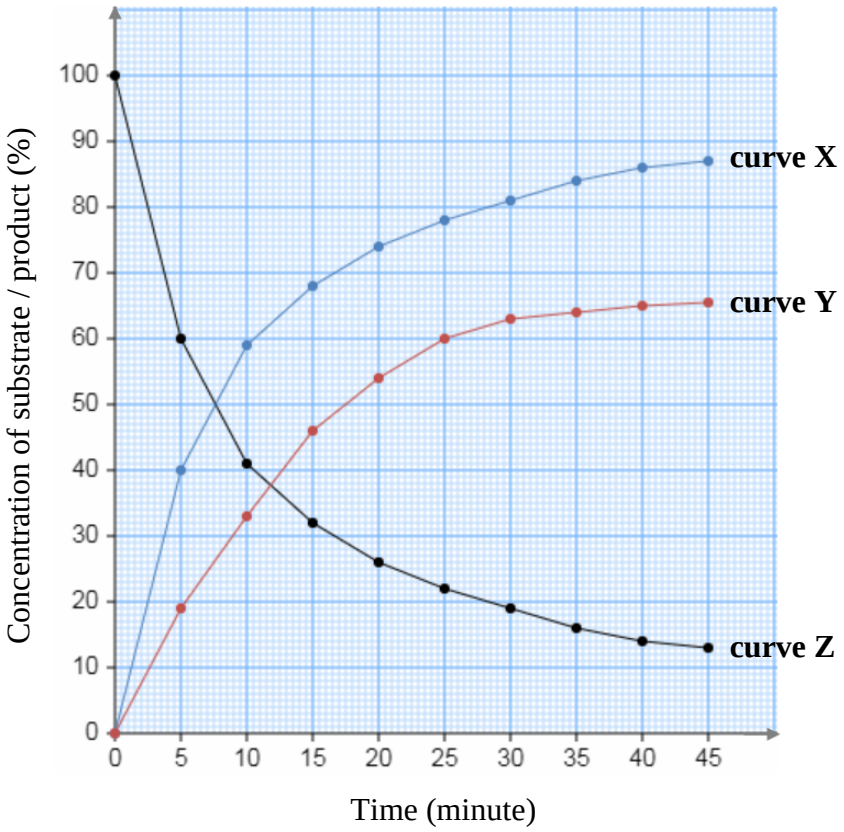
Structured questions

|!|BCELA0050000001|!| ★★ SI

An experiment was carried out to investigate an enzyme-catalysed reaction. The concentration of a substrate or product was monitored in three tubes of reaction mixtures as shown in the graph below.



(a) Identify whether each of the three curves represents the changes in the concentration of a substrate or product. (3 marks)

Curve X: _____

Curve Y: _____

Curve Z: _____

(b) The reaction mixture in one of the three tubes was wrongly prepared, leading to a different reaction rate.

(i) Suggest with a reason which curve corresponds to the wrongly prepared tube. (2 marks)

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(ii) Suggest ***two*** possible errors during the preparation of the reaction mixture. (2 marks)

(c) Curve X eventually leveled off and did not reach 100%. Suggest an explanation for this phenomenon. (2 marks)

##

(a) Curve X: product (1)
Curve Y: product (1)
Curve Z: substrate (1)

(b) (i) Curve Y, (1)
because its reaction rate is lower than that of curve X and Z. (1)

(ii) Any ***two*** of the following: (2)

- Fewer enzymes were added. (1)
- Fewer substrates were added. (1)
- The reaction mixture may be contaminated by inhibitors. (1)
- A buffer solution with an unsuitable pH was added. (1)

(Or other reasonable answers)

(c) The reaction is reversible. (1)
As the products accumulate, the backward reaction occurs. (1)

(9 marks)

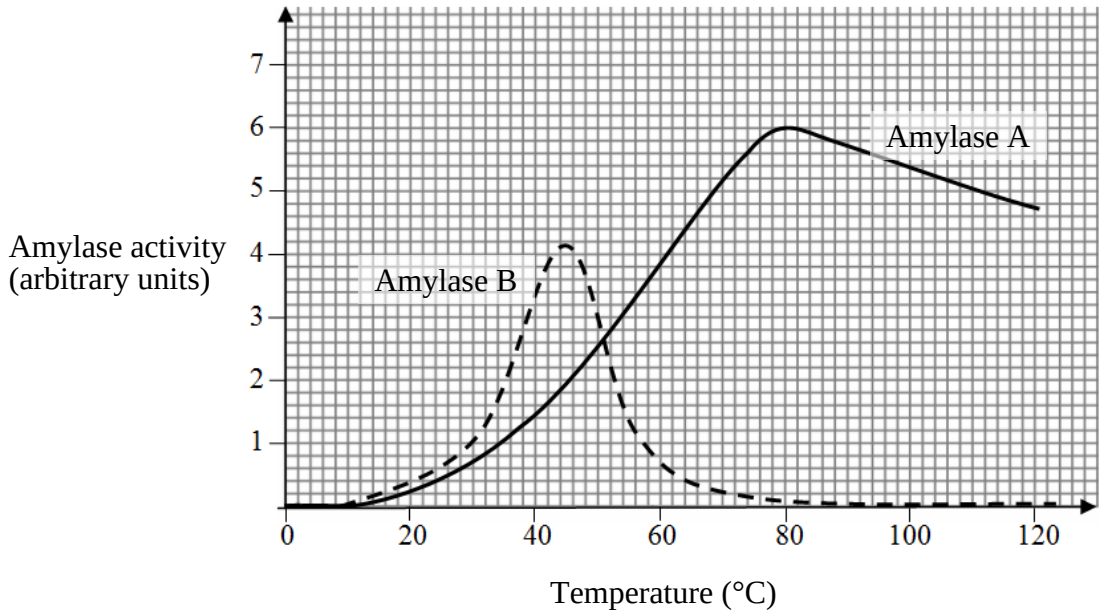
##

	Skills assessed	Marks
(a)	Analysing data	3
(b)(i)	Analysing data	1
	Applying concepts	1
(b)(ii)	Applying concepts	2
(c)	Applying concepts	2

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|!|BCELB0050000002|!| ★ **STSE**

The graph below shows the activity of two types of amylases. Amylase A is found in bacteria living in hot spring, while amylase B is found in barely grains.



(a) Describe the effect of increasing the temperature on the activity of amylase A. (4 marks)

(b) Amylase A is used in the industrial production of beer. It is mixed with germinated barley grains during the process.

(i) What is the action of amylase on the starch in barley grains? (1 mark)

(ii) Barley grains naturally contain amylase B, what are the advantages of adding additional amylase A to the barley grains? (2 marks)

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(iii) A batch of barley grains have been mixed with amylase A, but the grains have to be stored for use five days later. What temperature should the grains be stored at? Explain your answer. (3 marks)

##

(a) When the temperature is below 10 °C, there is no enzyme activity. (1)
When the temperature increases from 10 °C to 80 °C, the amylase activity increases. (1)
The optimum temperature for the activity of amylase is 80 °C. (1)
When the temperature is above 80 °C, the amylase activity decreases. (1)

(b) (i) Amylase catalyses the breakdown of starch in the grains into maltose. (1)

(ii) Allows a wider range of working temperature. (1)
The process can be carried out at a higher temperature. This shortens the time for the starch conversion process. (1)

(iii) The grains should be kept at below 10 °C. (1)
Both amylases A and B are inactive at this temperature (1)
but they are not denatured. (1)

(10 marks)

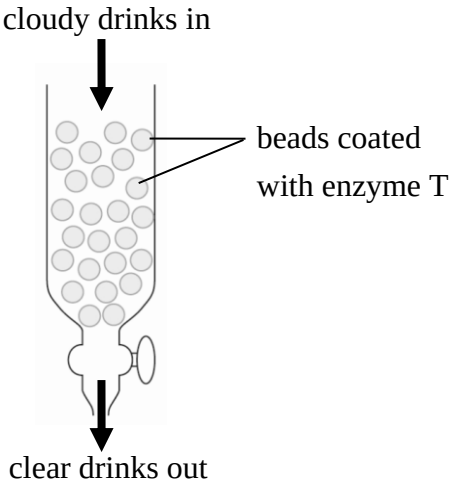
##

	Skills assessed	Marks
(a)	Analysing data	4
(b)(i)	Understanding basic concepts	1
(b)(ii)	Applying concepts	2
(b)(iii)	Applying concepts	3

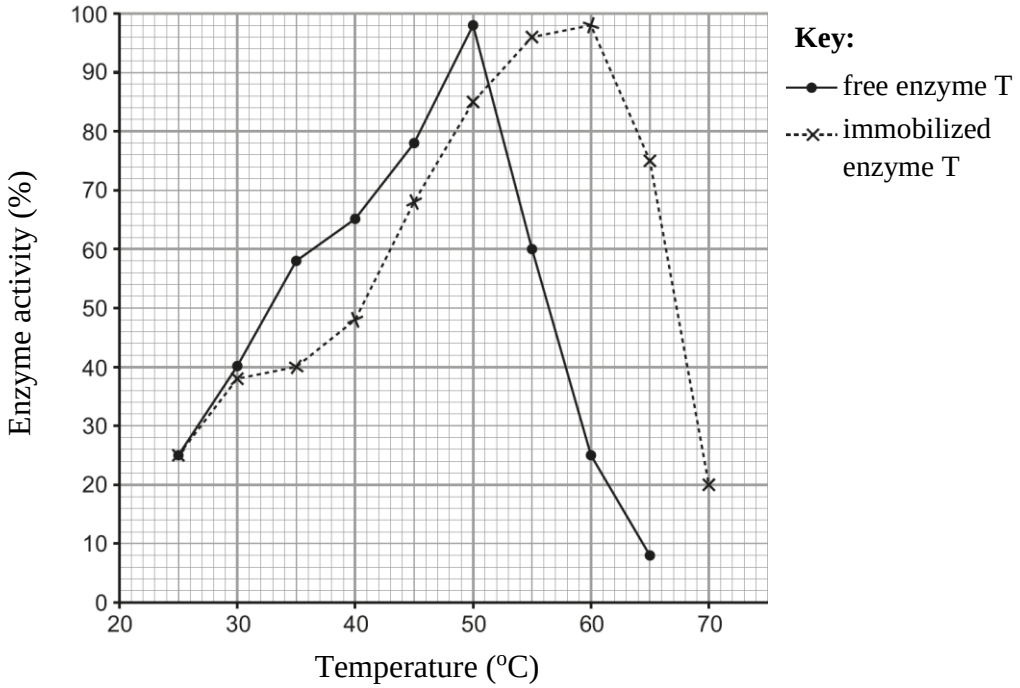
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|!|BCELA0050000003|!| ★★

An enzyme T is used commercially to clarify the cloudy appearance of drink products. This enzyme is often immobilized on the surfaces of small beads. When drink products are run through the spaces between the beads, the enzymes catalyse reactions to clarify the drinks.



The graph below shows the activity of equal concentrations of free enzyme T in solution and immobilized enzyme T over a range of temperatures.



(a) Enzyme T works both intracellularly and extracellularly. Explain what this means.

(2 marks)

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(b) Compare the activity of free enzyme T and immobilized enzyme T up to 50 °C. Explain the difference. (2 marks)

(c) With reference to the graph, describe the advantages of using immobilized enzyme T over free enzyme T. (3 marks)

##

(a) Enzyme T can exert functions inside cells (1)
and outside cells. (1)

(b) Below 50 °C, the activity of immobilized enzyme T is lower than that of free enzyme T. (1)
As enzyme T is fixed on beads, the chance of enzyme-substrate complex formation is smaller compared with free enzymes. (1)

(c) The optimum temperature of immobilized enzyme T is higher than that of free enzyme T. (1)
The clarifying process can be run at a higher temperature without making the enzyme denatured, (1)
so the process can complete in shorter time. (1)

(7 marks)

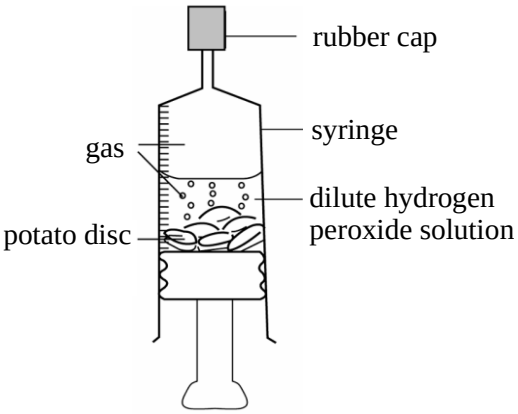
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	1
	Applying concepts	1
(c)	Analysing data	1
	Applying concepts	2

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|!|BCELB0050000004|!| ★ **SI**

Catalase is an enzyme that catalyses the breakdown of hydrogen peroxide to water and oxygen. An experiment was carried out to study the activity of catalase in potato tubers. A potato cylinder of 4 cm long was cut from a potato tuber using a cork borer. The cylinder was then cut evenly into 10 discs. The potato discs were then put into the set-up shown below.



The volume of gas collected in the syringe was recorded throughout the experiment as shown in the table below.

Time (minute)	**Volume of gas collected (cm^3)**
0	0
1	1.2
2	2.2
3	2.7
5	3.2
10	4.3
15	4.8
20	5.2
25	5.3
30	5.3

(a) Present the results of the experiment in the form of a graph. (4 marks)

(b) Describe and explain the changes in the reaction rate throughout the experiment. (4 marks)

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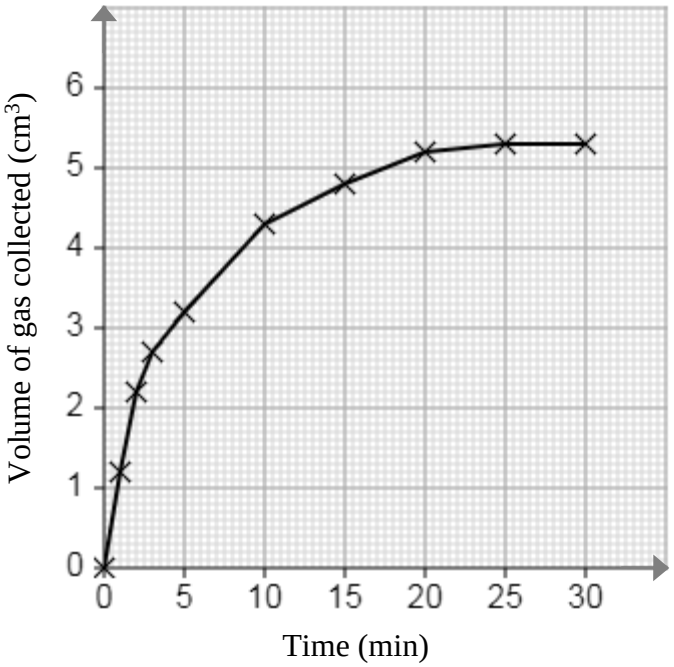
(c) If the 4 cm long potato cylinder was cut evenly into 20 discs instead of 10, how would the reaction rate change? Sketch a curve on the same graph that you plotted in (a) to illustrate this change. (2 marks)

_____

_____

##

(a)



Correct title (1)
Correct choice of axes (1)
Axes with labels and units (1)
Correct plotting and joining of line (1)

(b) As the reaction proceeded, the reaction rate decreased. (1)
As the reaction proceeded, the substrate concentration decreased. (1)
Therefore, the chance of collision between the substrate and enzyme molecules became smaller. (1)
When all the substrates were used up, the reaction stopped. (1)

(c) Sketching of a curve to the left of the one plotted in (a) (1)
The curve levels off at 5.3 cm^3 of gas collected (1)

(10 marks)

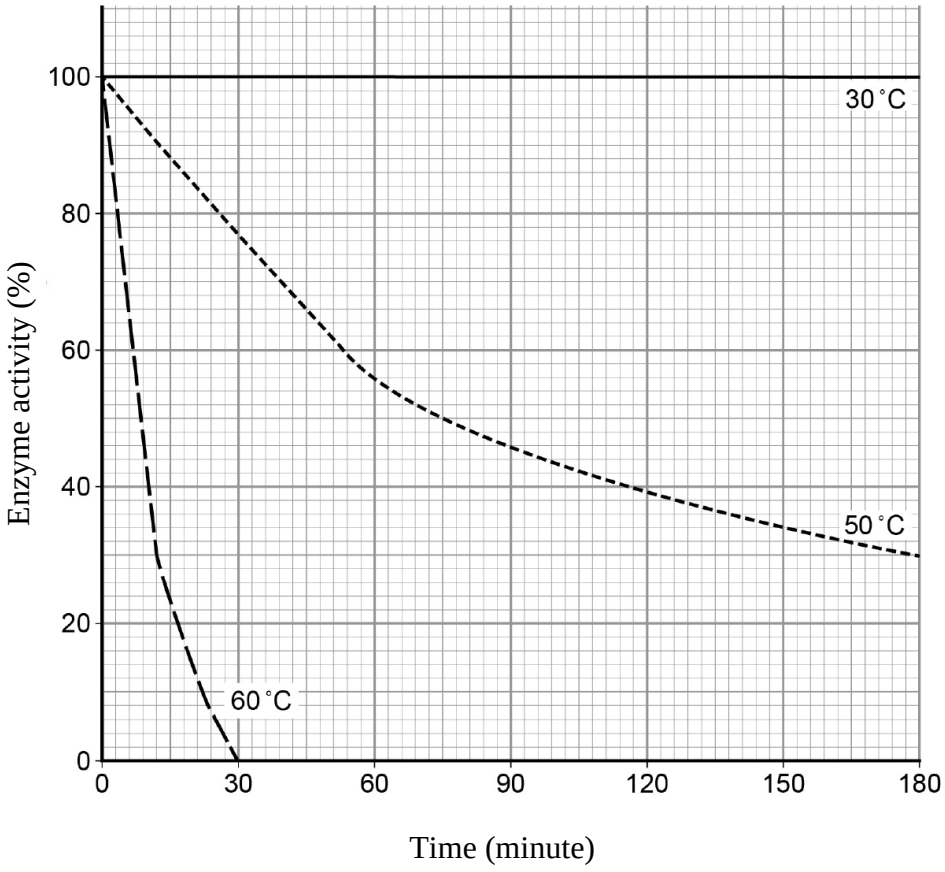
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##

	Skills assessed	Marks
(a)	Presenting information	4
(b)	Analysing data	1
	Applying concepts	3
(c)	Presenting information	1
	Applying concepts	1

|!|BCELA0050000005|!| ★★ **STSE**

The graph below shows the effect of temperature on the activity of a lipase used in biological washing powder.



(a) Enzymes are biological catalysts. Describe what is meant by the term biological catalyst. (2 marks)

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(b) Industrial enzymes are often extracted from bacteria. Suggest an advantage of this. (2 marks)

(c) Describe and explain the enzyme activities at 30 °C, 50 °C and 60 °C respectively. (6 marks)

(d) Biological washing powders are effective in removing blood stains and food stains from clothes. Suggest why biological washing powders often contain multiple types of enzymes. (2 marks)

##

(a) Enzymes are protein molecules produced by cells. (1)
They speed up the rate of reactions without being used up or changed themselves. (1)

(b) Bacteria divide rapidly. (1)
A large number of bacteria can be grown in a short time. (1)

(c) Enzyme activity remains high at 30 °C. (1)
30 °C is the optimum temperature for this enzyme. (1)
Enzyme activity decreases at 50 °C and 60 °C. (1)
The high temperatures cause denaturation of the enzymes. (1)
Denaturation is faster at 60 °C (1)
due to higher kinetic energy which break more bonds that hold the shape of the active sites of enzyme molecules. (1)

(d) Enzymes are specific in action. (1)
Blood stains and egg stains need proteases to break them down, while grease needs lipases to remove. (1)

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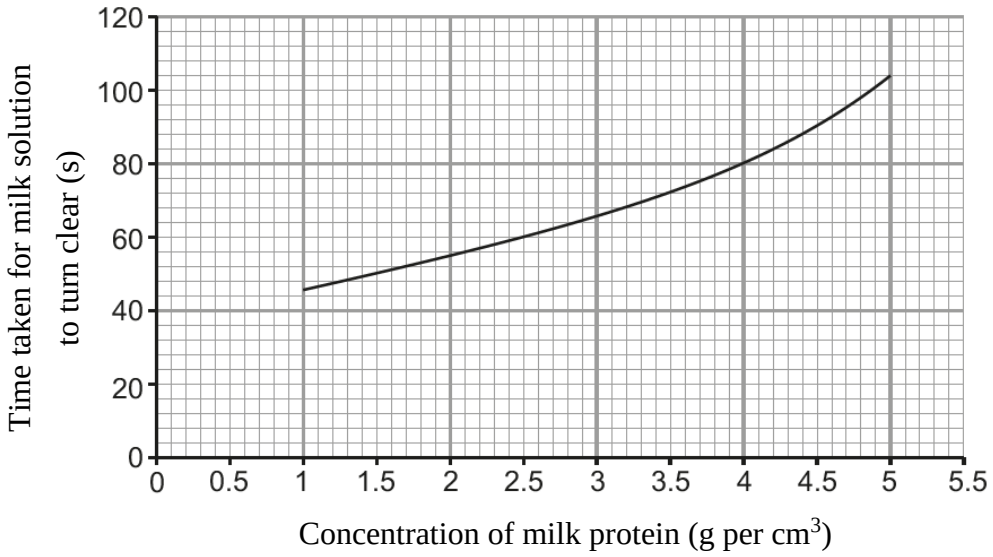
(12 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Applying concepts	2
(c)	Analysing data	3
	Applying concepts	3
(d)	Applying concepts	2

|!|BCELB0050000006|!| ★ **Cross topic** **SI**

An enzyme X can be used to digest proteins in milk. Milk mixed with enzyme X will turn clear when the proteins are broken down. The protein content of a milk solution can be determined by measuring the time taken for the solution to turn clear. A standard curve for the digestion of milk proteins using enzyme X is shown below.



(a) Two unknown samples of milk were tested. Sample P took 70 seconds to turn clear, while sample Q took 48 seconds. Determine the concentration of milk proteins in these two samples. (2 marks)

(b) How would you conduct a simple test to check whether the clear solution resulted contains proteins? (2 marks)

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The table below shows the concentration of proteins in milk produced by different types of mammals.

Mammal	Concentration of proteins in milk (g per cm^3)	Time for babies to double its mass (days)
Human	1.2	180
Cow	3.3	47
Goat	4.2	19

(c) With reference to the information given, suggest from which mammals sample P and Q were taken. (2 marks)

(d) Newborn mammals feed on the milk produced by their mothers. Give ***two*** advantages of breast-feeding to infants. (2 marks)

(e) Describe the pattern shown by the data in the table. Suggest an advantage for this. (2 marks)

##

(a) Sample P: 3.3 g per cm^3 (1)
Sample Q: 1.2 g per cm^3 (1)

(b) Dip a protein test strip into the sample. (1)
If protein is present, a distinctive colour change can be observed at the test end. (1)

(c) Sample P: cow (1)
Sample Q: human (1)

(d) Any ***two*** of the following: (2)
- Breast milk contains the right amounts of all the essential nutrients for the baby. (1)
- Breast milk is easy for babies to digest and absorb. (1)
- Breast milk contains antibodies that help protect babies against some infectious diseases, such as pneumonia, ear infections, etc. (1)

(e) The higher the protein content of the milk, the shorter the time taken for babies to double body mass. (1)
Babies can grow faster and have a greater chance of survival. (1)

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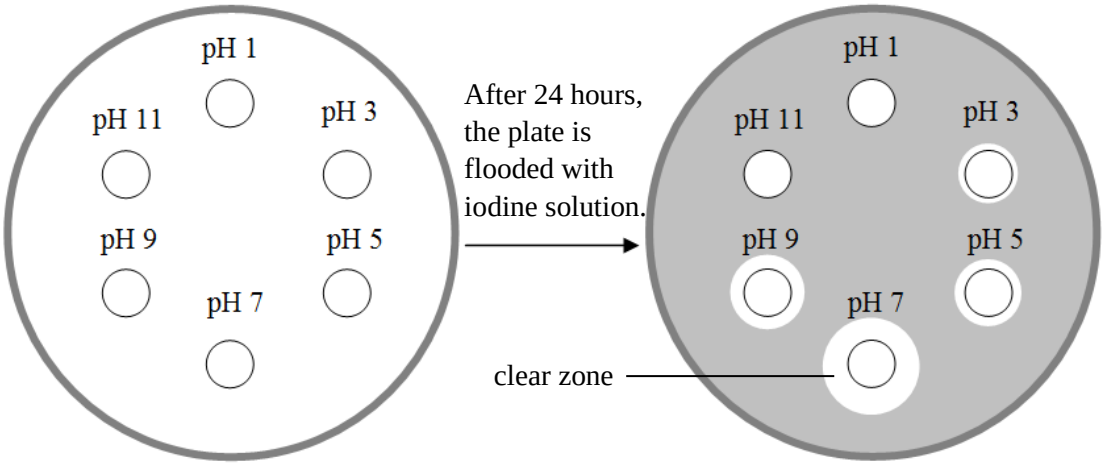
(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Practical skills	2
(c)	Analysing data	2
(d)	Understanding basic concepts	2
(e)	Analysing data	1
	Applying concepts	1

|!|BCELB0050000007|!| ★ **SI**

The enzyme amylase breaks down starch. A student carried out an investigation to study the effect of pH on amylase activity using a starch agar plate. Six wells were created in the agar plate using a cork borer. Each well contained the same volume and concentration of amylase, and a buffer solution of different pH. The agar plate was then left in the incubator at 40 °C for 24 hours.



(a) How could the student use these results to compare the activity of amylase at different pH values? (2 marks)

(b) The student concluded that the optimum pH for amylase activity was 7. Explain why this conclusion may not be valid. (1 mark)

(c) Explain the result obtained at pH 11. (2 marks)

(d) Design a control for this experiment. (2 marks)

##

(a) Use a piece of graph paper to estimate the area of each clear zone / use a ruler to measure the diameter of each clear zone. (1)
Compare the areas / diameters of the clear zones. (1)

(b) The experiment should be repeated at pH values from 6 to 8 in order to find the optimum pH for the highest amylase activity. (1)

(c) The enzymes denatured at pH 11. (1)
Substrates cannot bind to the active sites. (1)

(d) Use boiled amylase and (1)
carry out the experiment at all pH values. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Practical skills	1
(c)	Analysing data	2
(d)	Practical skills	2

|!|BCELB0050000008|!| ★ SI

The enzyme sucrase can speed up the digestion of sucrose to give glucose and fructose.

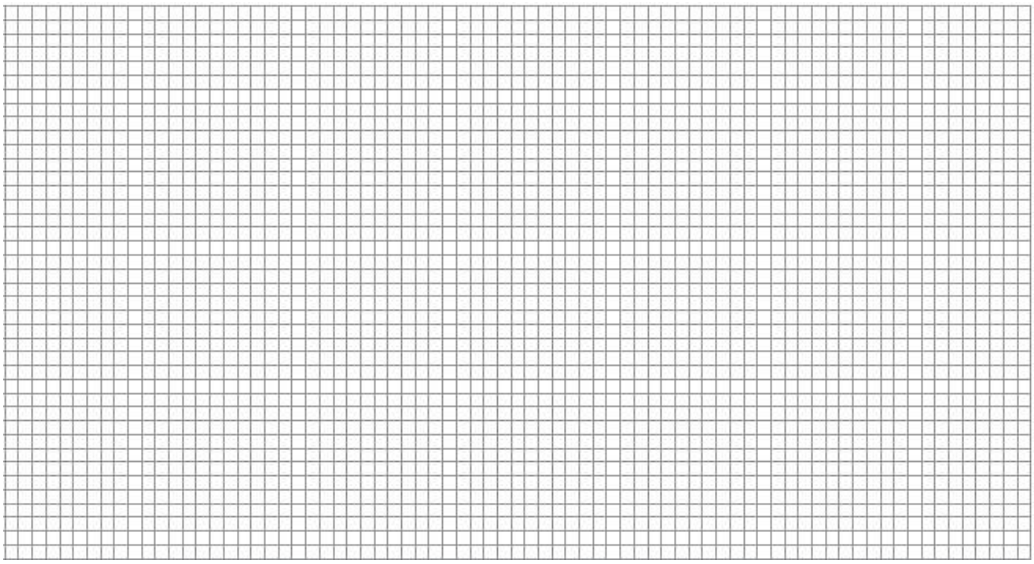
$$\text{sucrose} \xrightarrow{\text{sucrase}} \text{glucose} + \text{fructose}$$

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A student carried out an experiment to study the effect of increasing the concentration of sucrase on the rate of this reaction. He kept the concentration of sucrose constant. He used seven different concentrations of sucrase. For each of the set-ups, he measured the time taken for the sucrose to be completely digested. All the reactions were carried out at 40 ^{o}C. The results are shown in the table below.

Sucrase concentration (%)	**Time required to digest the sucrose (seconds)**
0.2	560
0.4	420
0.6	320
0.8	220
1.0	150
1.2	120
1.4	100

(a) Plot a graph to show the effect of sucrase concentration on the rate of sucrose digestion. (4 marks)



(b) With reference to your graph, describe the effect of increasing the concentration of sucrase on the rate of sucrose digestion. (2 marks)

_____

_____

(c) Explain why all the reactions were carried out at 40 ^{o}C. (1 mark)

_____

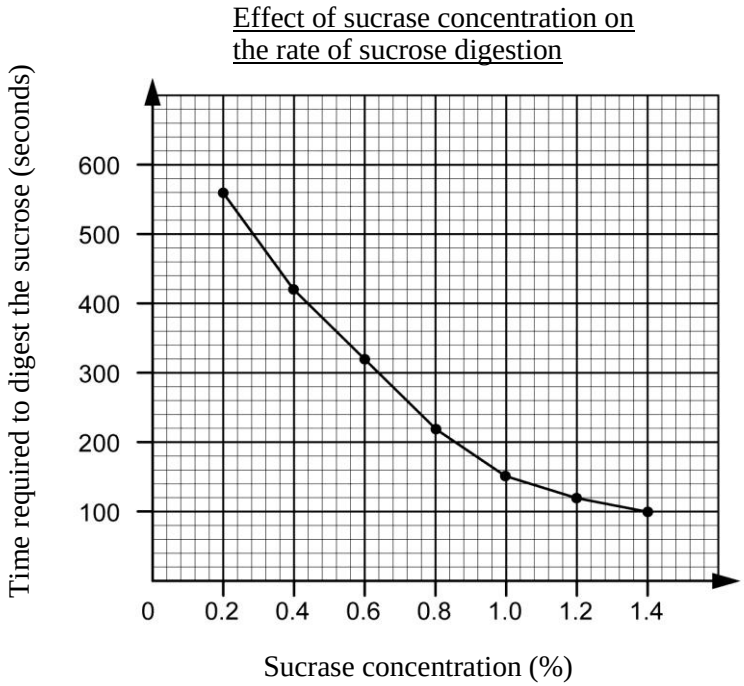
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(d) Describe how the temperature can be kept constant at 40 °C. (1 mark)

(e) What would be the results if the experiment were carried out at 80 °C? Explain. (2 marks)

##

(a)



Correct title (1)
Correct choice of axes (1)
Axes with labels and units (1)
Correct plotting and joining of line (1)

(b) The time required to digest sucrose becomes shorter (1)
when the concentration of sucrase increases. (1)

(c) Because this is the optimum temperature for sucrase to work / to minimize the effect of temperature on the activity of sucrase. (1)

(d) Put all set-ups into a water bath at 40 °C (1)

(e) Sucrose would still exist after a long period of time (1)
because sucrase was denatured at high temperatures. (1)

(10 marks)

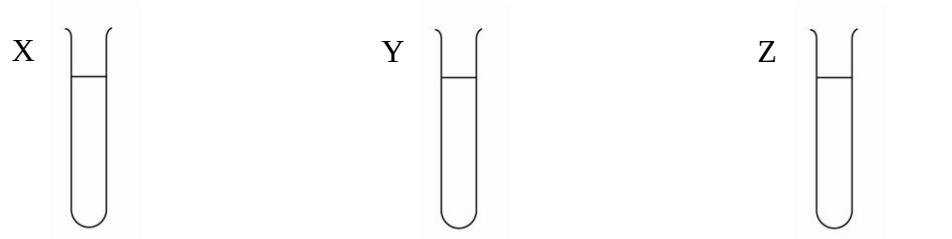
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##

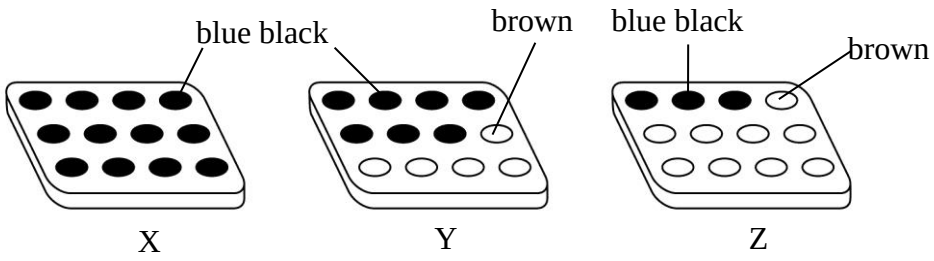
	Skills assessed	Marks
(a)	Presenting information	4
(b)	Analysing data	2
(c)	Applying concepts	1
(d)	Understanding basic concepts	1
(e)	Applying concepts	2

|!|BCELC0050000009|!| **SI**

An experiment was carried out to investigate the effect of amylase on starch.



Samples were taken from each tube every minute and added to iodine solution on a spotting tile.



(a) What effect does amylase have on starch? Give evidence for this. (3 marks)

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(b) What is the effect of temperature on the action of amylase? Give evidence for this. (2 marks)

(c) What is the product when amylase is mixed with starch? How can you show its presence? (3 marks)

##

(a) Amylase catalyses the breakdown of starch. (1)
In X, amylase is absent and starch still exists. (1)
In Y and Z, amylase is present and starch disappears after a certain period of time. (1)

(b) Amylase works faster at higher temperature. (1)
In Z, the temperature is kept at 40 ^{o}C and the starch disappears faster than at 20 ^{o}C. (1)

(c) It is maltose / reducing sugar. (1)
Heat the products with Benedict's solution. (1)
If maltose / reducing sugar is present, orange / brick-red precipitates appear. (1)

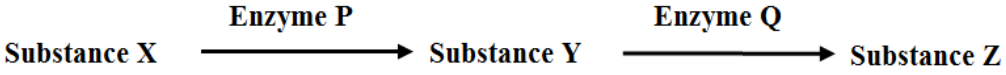
(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Analysing data	2
(c)	Understanding basic concepts	3

|!|BCELC0050000010|!|

The diagram below shows a biochemical pathway.



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(a) What is enzyme P's

(i) substrate? (1 mark)

(ii) product? (1 mark)

(b) What is enzyme Q's

(i) substrate? (1 mark)

(ii) product? (1 mark)

(c) Enzyme P is affected by an inhibitor. If equal concentrations of substances X, Y and Z are put into a container containing enzymes P and Q and a high concentration of the inhibitor is added, what will happen to the concentrations of substances X, Y and Z? (3 marks)

##

(a) (i) Substance X (1)
(ii) Substance Y (1)
(b) (i) Substance Y (1)
(ii) Substance Z (1)
(c) The concentration of substance X will remain unchanged. (1)
The concentration of substance Y will decrease and (1)
that of Z will increase. (1)

(7 marks)

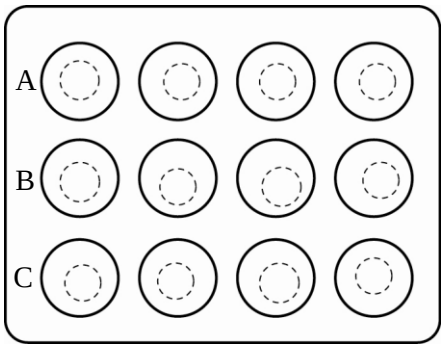
##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(c)	Applying concepts	3

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|!|BCELA0050000011|!| ★★ **SI**

Phosphorylase is an enzyme catalysing the synthesis of starch from glucose-1-phosphate. In an investigation to study the action of phosphorylase, drops of phosphorylase, glucose-1-phosphate and distilled water were added into the wells of a spot plate as shown in ***Figure I***.



Row A: phosphorylase + glucose-1-phosphate

Row B: phosphorylase + distilled water

Row C: glucose-1-phosphate + distilled water

Figure I

At three-minute intervals, a drop of iodine solution was added to one well in each row. ***Figure II*** shows the results.

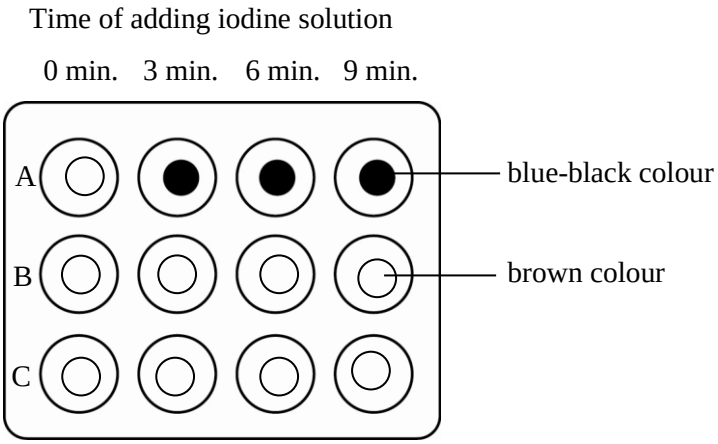


Figure II

(a) Explain the formation of blue-black colour after adding iodine solution to row A. (1 mark)

_____

(b) The above experiment was carried out at 25 °C. What would be the change in the results in row A if the experiment was carried out at 0 °C? Explain your answer. (4 marks)

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(c) What are the functions of setting up rows B and C in this experiment? (2 marks)

(d) What conclusion can be drawn from the results of the experiment? (1 mark)

(e) Glucose-1-phosphate can be found in potato tubers. Based on the results of the experiment, deduce the role of phosphorylase in the development of potato tubers. (2 marks)

##

(a) The presence of starch changes iodine solution from brown to blue-black colour. (1)

(b) It would take a longer time for the blue-black colour to appear. (1)
Low temperature decreases the kinetic energy of the enzyme and substrates. (1)
There would be a smaller chance of collisions between enzyme and substrate, (1)
so fewer enzyme-substrate complexes would form. (1)

(c) Row B can be used to show that the synthesis of starch requires glucose-1-phosphate. (1)
Row C can be used to show that the synthesis of starch requires phosphorylase. (1)

(d) The presence of phosphorylase and glucose-1-phosphate can lead to the production of starch.
OR
Phosphorylase and glucose-1-phosphate are needed for the production of starch. (1)

(e) Phosphorylase can convert the glucose-1-phosphate in potato cells to starch. (1)
Starch is readily stored in the potato tuber as food reserve. (1)

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(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	4
(c)	Practical skills	2
(d)	Analysing data	1
(e)	Applying concepts	2

|!|BCELB0050000012|!| ★★ **STSE**

Biological washing powder contains enzymes.

(a) Suggest ***two*** types of enzymes that may be present in biological washing powder. (2 marks)

(b) Give ***one*** advantage and ***one*** disadvantage of adding enzymes into washing powder. (2 marks)

(c) (i) Name a factor that would greatly affect the effectiveness of biological washing powder. (1 mark)

(ii) Based on your answer in (i), suggest ***one*** way to ensure the proper working of biological washing powder when washing clothes. (1 mark)

(d) The enzymes in biological washing powder are usually packed inside small granules, and are only released when they come into contact with water. Suggest ***one*** advantage of this. (1 mark)

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##

(a) Any ***two*** of the following: (2)

protease / lipase / amylase (1 mark each)

(b) Any ***one*** of the following advantage: (1)

- Enzymes speed up the breakdown of proteins, lipids and starch in the food stains on clothes.
- It can remove food stains at a lower temperature compared with non-biological washing powder.

(Or other reasonable answer)

Any ***one*** of the following disadvantage: (1)

- Protease and lipase may damage the skin during hand-washing.
- Protease may damage some fabrics.

(Or other reasonable answer)

(c) (i) Factor: water temperature (1)

(ii) Do not use hot water / Use warm water to wash clothes. (1)

(d) It causes less harm to the users. (1)

(7 marks)

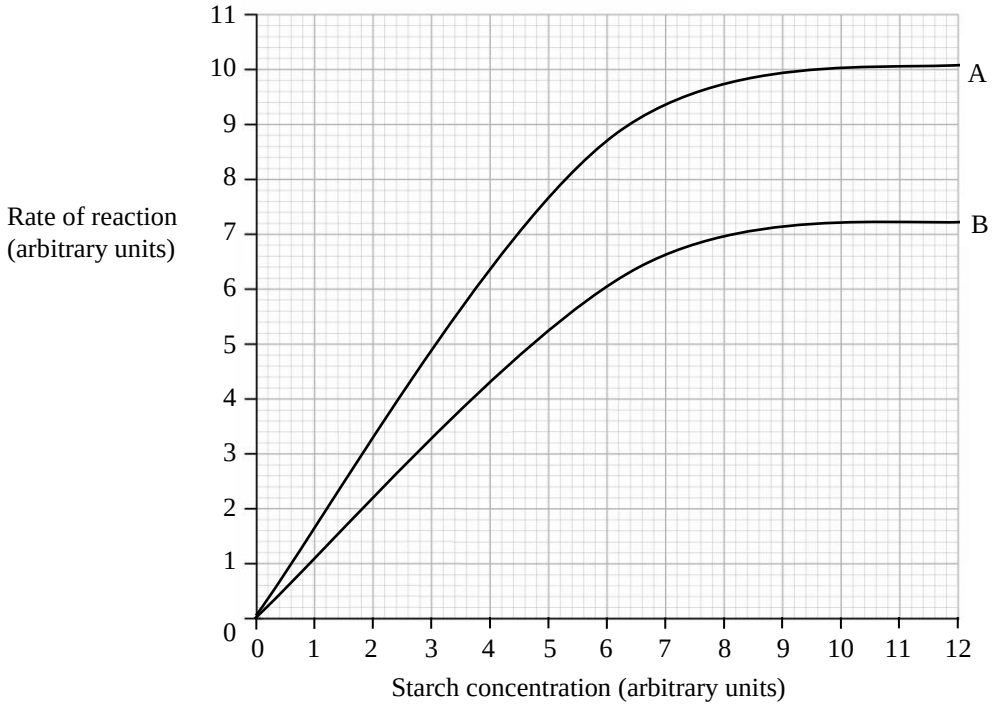
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)(i)	Applying concepts	1
(c)(ii)	Applying concepts	1
(d)	Applying concepts	1

|!|BCELB0050000013|!| ★

In an experiment, the rate of starch breakdown as catalysed by amylase was measured in solutions with different starch concentrations. The experiment was carried out at 25 ºC and was repeated with mercury ions added to the starch solutions. The following graph shows the results.

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(a) What are the products of the reaction catalysed by amylase? (1 mark)

(b) Which curve represents the results of the set-up with mercury ions added? Explain your answer. (3 marks)

(c) Why do both curves level off beyond starch concentration of 9 arbitrary units? (2 marks)

(d) Sketch on the graph to show the change in the reaction rate of amylase when the experiment was carried out at 37 ºC in the absence of mercury ions. (1 mark)

##

(a) Maltose (1)

(b) Curve B (1)

Mercury ions inhibit amylase from breaking down starch. (1)

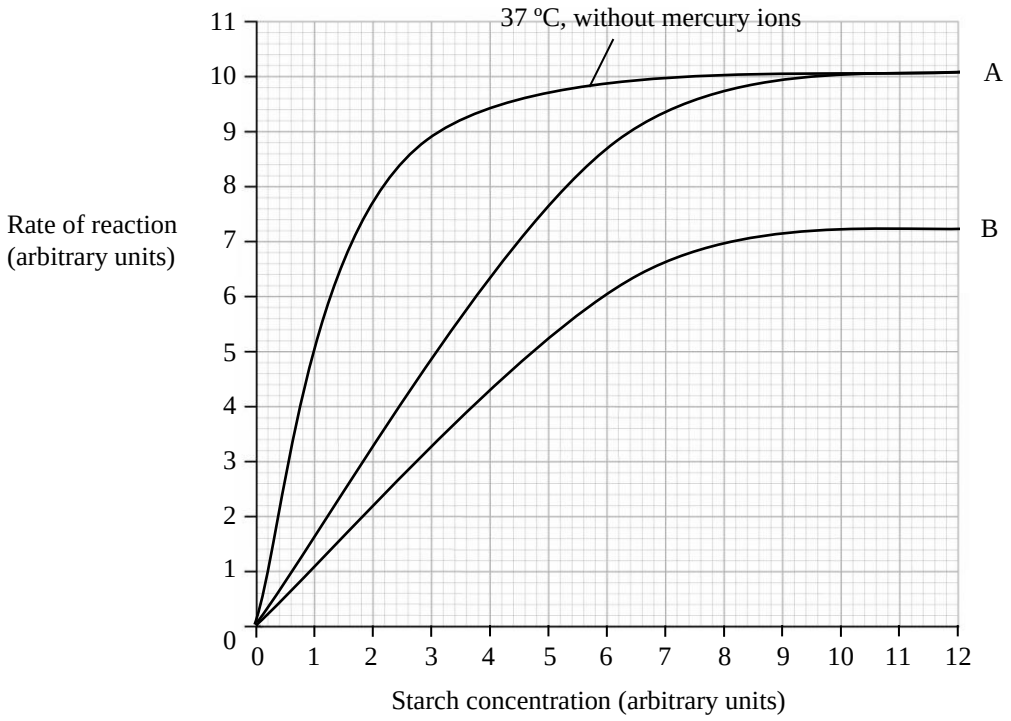
The rate of the reaction is thus lowered. (1)

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(c) At the starch concentration above 9 arbitrary units, enzyme concentration becomes the limiting factor / substrate is no longer limiting. (1)

All the active sites of enzymes are saturated with substrates. (1)

(d) Correct drawing of curve (1)



(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Analysing data	1
	Applying concepts	2
(c)	Applying concepts	2
(d)	Presenting information	1

|!|BCELB0050000014|!| ★ **STSE**

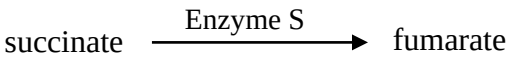
During sterilization, we can kill bacteria by interfering with their enzyme activities.

(a) (i) Most bacteria are temperature and pH sensitive. According to this property of bacteria, suggest how we can kill these bacteria. (2 marks)

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(ii) Explain the mechanisms behind the methods in (a)(i). (3 marks)

(b) Besides the methods mentioned in (a)(i), we can kill bacteria by inhibiting the enzymes essential in some important metabolic pathways. The following shows a reaction in the process of respiration in bacteria.



A malonate molecule is structurally similar to a succinate molecule. A malonate molecule can thus bind to the active site of enzyme S.

Suggest how malonate can be used to kill bacteria. (4 marks)

##

(a) (i) Heat sterilization / Use of high temperature (1)
Use of strong acid / strong base (1)
(ii) High temperature or extreme pH alters the shapes of the active sites of enzymes in the bacteria. (1)
This causes denaturation of enzymes and the enzymes lose their catalytic functions. (1)
The metabolism of the bacteria is hindered and the bacteria may die. (1)

(b) Melonate molecules may occupy the active site of enzyme S. (1)
This prevents the formation of enzyme-substrate complexes and so the reaction rate decreases. (1)
Less fumarate is produced and the respiration process is inhibited. (1)
Bacteria may die due to lack of energy. (1)

(9 marks)

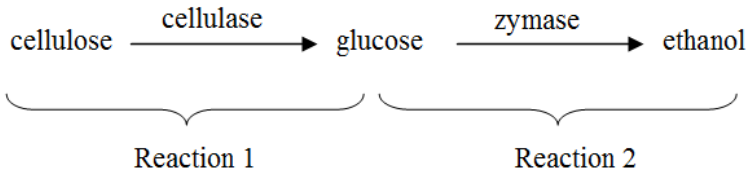
##

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	Skills assessed	Marks
(a)(i)	Understanding basic concepts	2
(a)(ii)	Applying concepts	3
(b)	Applying concepts	4

|!|BCELB0050000015|!| ★ **STSE**

In the production of biofuels, cellulose is first broken down into simple sugars with cellulase. Sugars are then fermented to produce ethanol, which can be used as biofuels. The following shows the two reactions in biofuel production.



(a) Suggest a source of cellulose for the production of biofuels. (1 mark)

(b) Explain how the rate of reaction 2 will be affected if the amount of cellulose increases. (2 marks)

(c) Heat is produced during the process of fermentation. Explain the effects of heat on reaction 2. (4 marks)

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(d) Suggest another industrial application of cellulase. (1 mark)

##

(a) Plant materials / crops / other reasonable answer (1)

(b) An increased amount of the substrate cellulose will lead to an increase in the amount of glucose produced. (1)

An increased amount of the substrate glucose will lead to an increase in the rate of reaction 2. (1)

(c) An increased temperature will increase the chance of enzyme-substrate complex formation. (1)

This will increase the rate of reaction 2 at first. (1)

As the temperature increases above the enzyme's optimum temperature, the shape of the active site of the enzymes changes and the enzymes lose their catalytic function. (1)

The rate of reaction 2 decreases as a result. (1)

(d) Making stonewashed jeans / fruit juice extraction / other reasonable answer (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	2
(c)	Applying concepts	4
(d)	Understanding basic concepts	1

|!|BCELB0050000016|!| ★ **STSE**

Read the passage below and answer the questions that follow.

Toxin in raw soy milk

Soy milk is a nutritious drink. However, homemade soymilk may be toxic if it is not heated properly. Soy beans contain an enzyme inhibitor, which inhibits the action of trypsin—a pancreatic protease. The trypsin inhibitor is a protein that binds to trypsin and changes the shape of its active site. This inhibits the digestion and absorption of proteins. It is advised to thoroughly boil soy milk during the production process.

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(a) Which medium (acidic, neutral or alkaline) is the most suitable for the action of trypsin? (1 mark)

(b) According to the passage, why is the trypsin inhibitor regarded as a 'toxin'? (3 marks)

(c) How can boiling minimize the effect of the trypsin inhibitor? (2 marks)

(d) The presence of the trypsin inhibitor is important to the soy plants in the wild. Suggest why. (1 mark)

(e) In terms of natural selection, suggest how soy plants have evolved the presence of the trypsin inhibitor in seeds. (4 marks)

##

(a) Alkaline (1)

(b) The trypsin inhibitor changes the shape of the active site of trypsin, so substrates can no longer fit to the active site. (1)
Without functioning trypsin, the digestion of protein into peptides and amino acids is hindered. (1)
Fewer amino acids are absorbed for growth and repair / synthesis of enzymes / other functions of proteins. (1)

(c) Trypsin inhibitor is protein in nature. (1)
Boiling provides a high temperature which leads to the denaturation of trypsin inhibitor. (1)

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(d) This can prevent herbivores from ingesting / consuming them. (1)

(e) Genetic variation exists among individuals of soy plants. Some individuals contain the trypsin inhibitor. (1)

These individuals can avoid the consumption by pests / herbivores. (1)

They reproduce and pass on the favourable characteristics to their offspring. (1)

Over many generations, the proportion of soy plants having the trypsin inhibitor increases. (1)

(11 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	3
(c)	Applying concepts	2
(d)	Applying concepts	1
(e)	Applying concepts	4

|!|BCELB0050000017|!| ★ **SI**

In an experiment to study the effect of temperature on the milk-coagulating action of an enzyme called rennin, milk and the enzyme were first kept in separate test tubes in water baths of different temperatures for 15 minutes before they were mixed. The resulting mixtures were kept in the same water bath. The time taken for the milk to coagulate was noted for each water bath of different temperature. The results are shown in the table below.

Temperature of water bath (°C)	Time taken for coagulation (minutes)
10	No coagulation occurred.
20	7
30	4
40	2
50	4
60	No coagulation occurred.

(a) What is the purpose of keeping the milk and enzyme in the water bath for 15 minutes before mixing? (1 mark)

_____

(b) At which temperature is the enzyme most active? (1 mark)

_____

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(c) State and explain whether there would be any change in the results if the mixture originally kept at 10 ºC was warmed to 40 ºC. (3 marks)

(d) How would you prepare a control set-up in order to confirm that the coagulation of milk is due to the action of an enzyme? What will be the expected results? (3 marks)

(e) State ***one*** factor, other than temperature, that can affect the activity of enzymes. (1 mark)

##

(a) To allow sufficient time for the milk and the enzyme to reach the required temperature. (1)

(b) 40 ºC (1)

(c) Coagulation will take place. (1)
This is because at 10 ºC, enzyme is inactive. (1)
However, enzyme will become active and functional again at 40 ºC. (1)
Thus coagulation will take place.

(d) Replace the enzyme solution with boiled enzyme solution / distilled water and repeat the experiment. (1)
The milk will not coagulate at all temperatures. (1)

(e) pH / enzyme concentration / substrate concentration / other reasonable answer (1)

(9 marks)

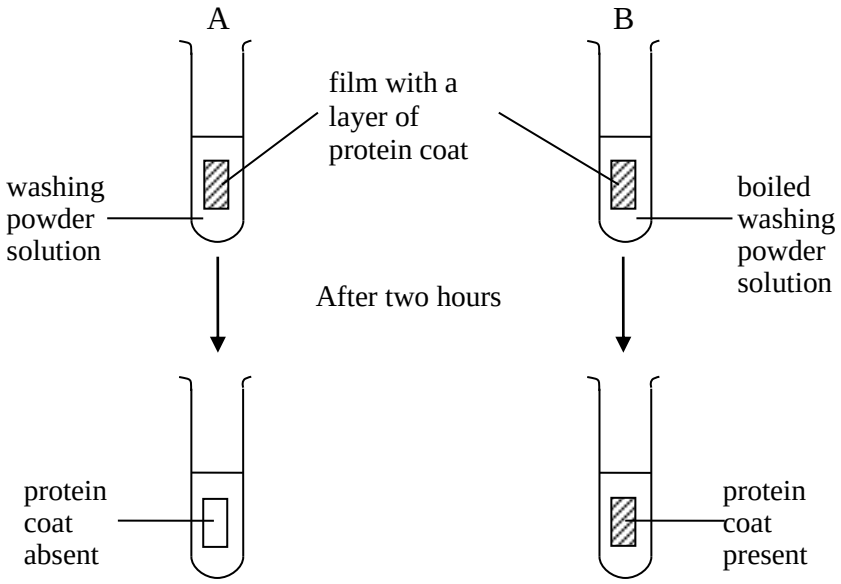
##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Analysing data	1
(c)	Applying concepts	3
(d)	Applying concepts	3
(e)	Understanding basic concepts	1

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|!|BCELB0050000018|!| ★ **SI** **STSE**

The following diagram shows an experiment that is used to study the properties of a biological washing powder containing a substance X.



(a) Describe the action of substance X on the protein coat. (1 mark)

_____

_____

(b) Name substance X. Explain why this washing powder can remove egg stains from clothes. (3 marks)

_____

_____

_____

(c) It is not advisable to use this washing powder with water above 60 ºC. Explain why. (1 mark)

_____

_____

(d) It is also suggested not to use this washing powder together with bleach solution. Explain briefly. (Hint: What is the pH of bleach?) (2 marks)

_____

_____

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##

(a) Substance X breaks down / digests the protein coat into amino acids. (1)

(b) X is an enzyme / protease. (1)

This washing powder contains enzyme / protease that digests the proteins in egg stains to amino acids (1)

which are soluble in water / can be removed by water. (1)

(c) The enzyme will be denatured at temperatures above 60 ºC. (1)

This decreases the effectiveness of the biological washing powder.

(d) Bleach is highly alkaline. (1)

The enzymes in the washing powder will become inactive or denatured at this high pH value. (1)

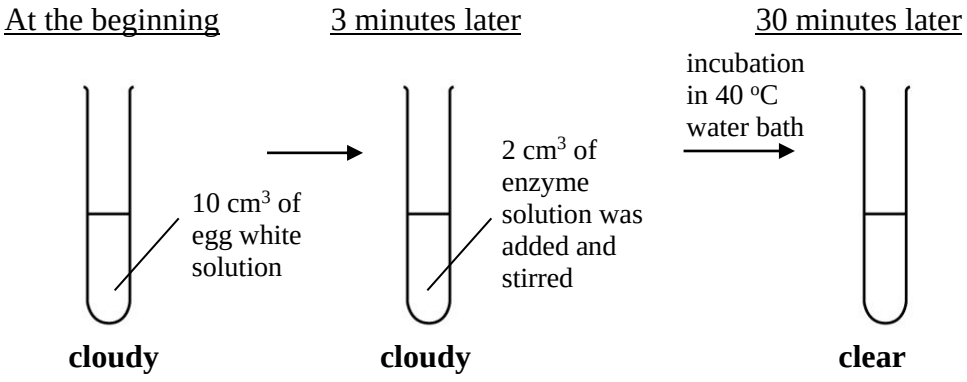
(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	3
(c)	Applying concepts	1
(d)	Applying concepts	2

|!|BCELC0050000019|!| **SI**

Tom carried out an experiment to investigate the action of an enzyme. The following diagram shows the procedure and the results of the experiment.



(a) Name the enzyme added to the egg white solution. (1 mark)

(b) Explain why the solution becomes clear. (1 mark)

(c) Suggest ***one*** method to speed up the action. (1 mark)

(d) Predict and explain the result if the test tube is placed in a water bath at

(i) 80 °C. (2 marks)

(ii) 2 °C. (2 marks)

##

(a) Protease (1)

(b) Protease breaks down the proteins of egg white into amino acids. (1)

(c) Increase the concentration of the enzyme / other reasonable answer (1)

(d) (i) The solution will remain cloudy because (1)
the enzyme will be denatured and cannot catalyse the breakdown reaction at such a high temperature. (1)

(ii) The solution will remain cloudy because (1)
the enzyme will become inactive and lead to a low reaction rate at low temperature. (1)

(7 marks)

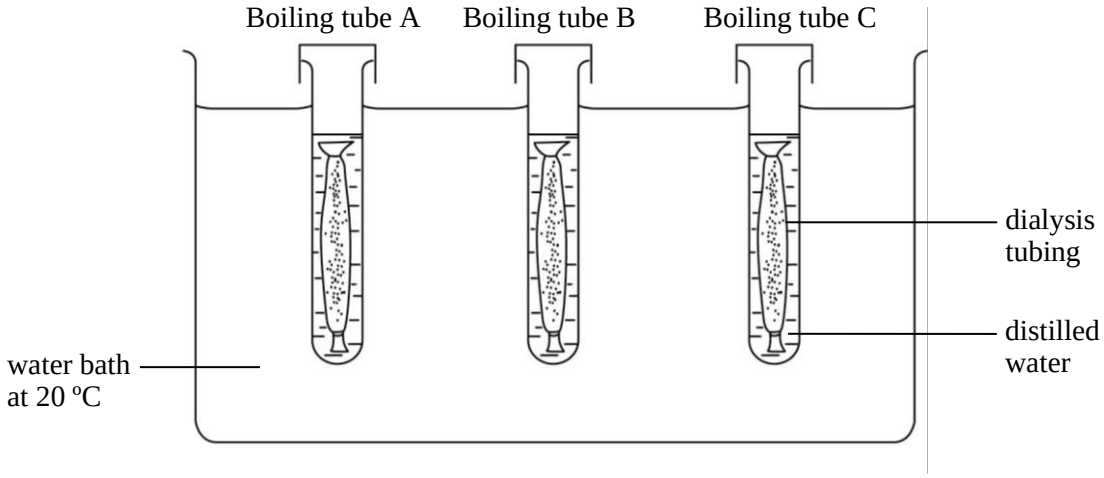
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)(i)	Applying concepts	2
(d)(ii)	Applying concepts	2

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|!|BCELB0050000020|!| ★ **SI**

The following shows an experimental set-up used to investigate the action of amylase on starch.



The following table shows the contents and the pH in each boiling tube.

Boiling tube	Contents inside the dialysis tubing	pH of the contents
A	starch + amylase	2
B	starch + amylase	7.5
C	starch + amylase	11

(a) State the action of amylase on starch. (2 marks)

_____

_____

(b) State the tube in which the action of amylase on starch is the fastest. Explain your answer. (2 marks)

_____

_____

(c) Design a control for this experiment. (1 mark)

_____

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(d) Suggest and explain a method to speed up the reaction. (2 marks)

(e) Predict, if the amylase can act on starch to form the end product, whether the distilled water surrounding the dialysis tubing contains the end product. Explain your answer. (2 marks)

##

(a) Amylase catalyses the reaction (1)
in which starch is broken down into maltose. (1)
(b) B, (1)
because amylase works best in slightly alkaline medium. (1)
(c) Use a similar set-up with the amylase replaced by distilled water. (1)
(d) The temperature of the water bath is kept at 40 ºC. (1)
This is the optimum temperature at which the enzyme is the most active. (1)
OR
A higher amylase concentration is used. (1)
Higher concentration of enzyme increases the rate of reaction. (1)
(e) Yes, (1)
because the end products inside the dialysis tubing are small enough to diffuse through the dialysis tubing to the distilled water outside. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	1
	Understanding basic concepts	1
(c)	Practical skills	1
(d)	Applying concepts	2
(e)	Applying concepts	2

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|!|BCELB0050000021|!| ★ SI

Read the passage below and answer the questions that follow.

Meat tenderization

Enzymes have been utilized by humans for thousands of years. One example is the use of enzymes by Africans. Much of the meat consumed in these areas is very fibrous and tough. This type of meat would need to be cooked for hours to break down the fibrous protein in order to make the meat more edible. The people in these areas have learned that after mixing meat with papaya extract for several hours, the meat becomes tenderer. Later, it is found that papaya contains an enzyme papain which is often used as meat tenderizers.

(a) What kind of tissue is found in meat? (1 mark)

(b) Name the substrate on which papain in the papaya extract works. (1 mark)

(c) Describe the action of the papain on the substrate. (2 marks)

(d) Explain why cooking stops meat tenderization by papain. (2 marks)

(e) Design an experiment to show the presence of protein-digesting enzymes in papaya extract. (3 marks)

##

(a) Muscle (1)

(b) The substrate is protein. (1)

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(c) The papain speeds up (1)
the breakdown of protein to amino acids. (1)

(d) Papain is denatured when cooked at high temperatures. (1)
The denatured enzyme cannot catalyse the breakdown process. (1)

(e) Add papaya extract, boiled papaya extract and distilled water to three cavities in a milk agar plate separately. (1)
Incubate the milk agar plate at 37 ^{o}C for two hours. (1)
Measure the size of clear zones around the cavities. (1)

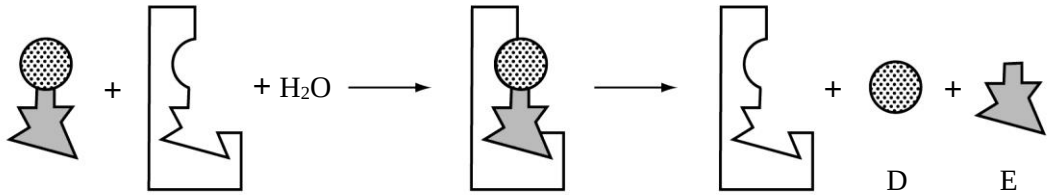
(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2
(d)	Applying concepts	2
(e)	Practical skills	3

|!|BCELC0050000022|!|

The diagram below shows how a certain chemical reaction takes place inside the small intestine of a mammal.



(a) Which of the above molecule is an enzyme? (1 mark)

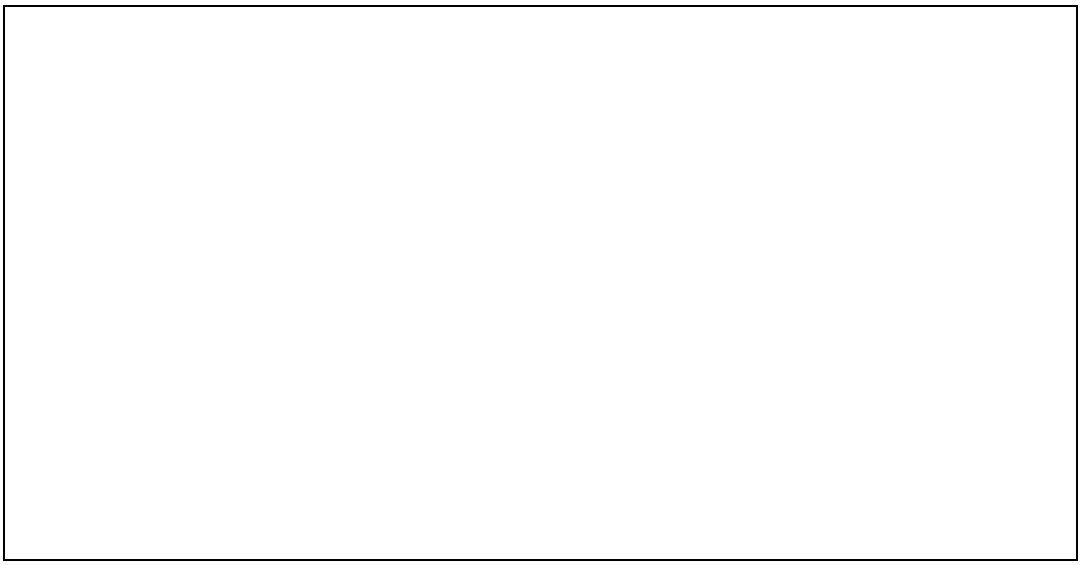
(b) Which of the above molecules are the substrate(s) and product(s) of this reaction? (2 marks)

(c) Which part of the enzyme determines its specificity? (1 mark)

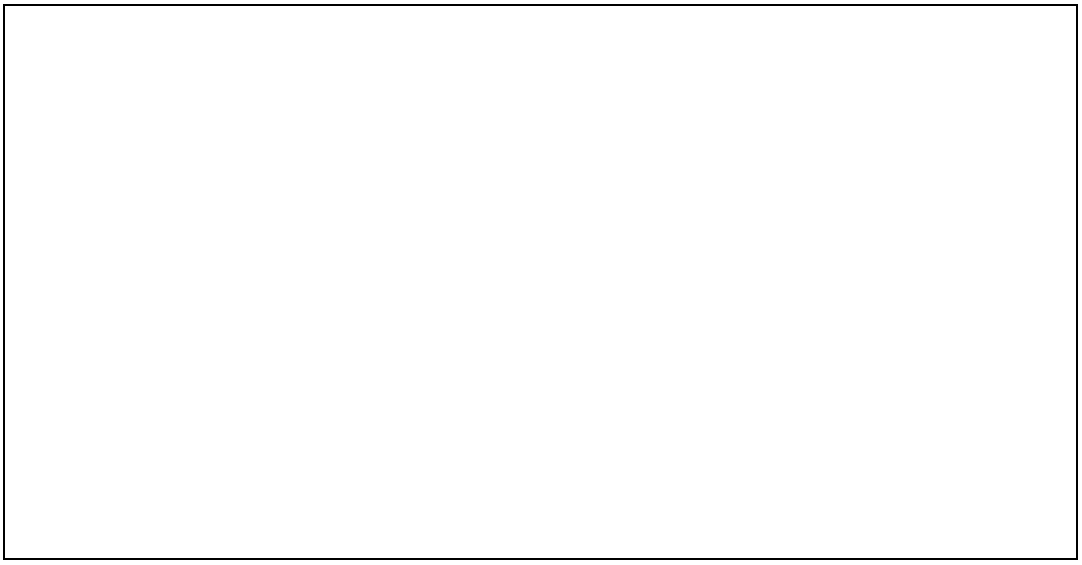
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(d) Sketch a graph to show the relationship between amylase activity and

(i) temperature. (2 marks)



(ii) pH. (2 marks)



##

(a) Molecule B (1)

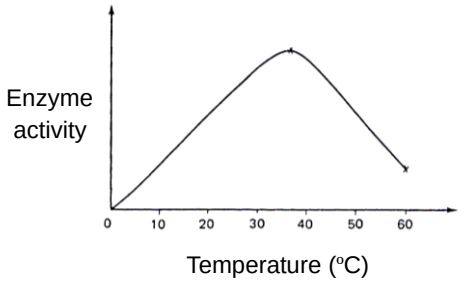
(b) Substrates: molecule A, water (1)

Products: molecules D and E (1)

(c) Active site (1)

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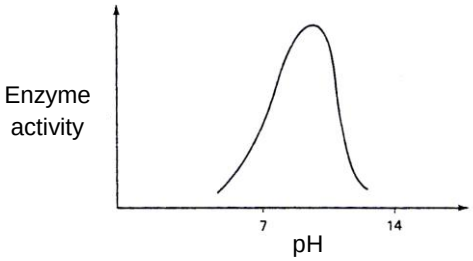
(d) (i)



Correct sketch of curve (1)

Correct labels of axes (1)

(ii)



Correct sketch of curve (1)

Correct labels of axes (1)

(8 marks)

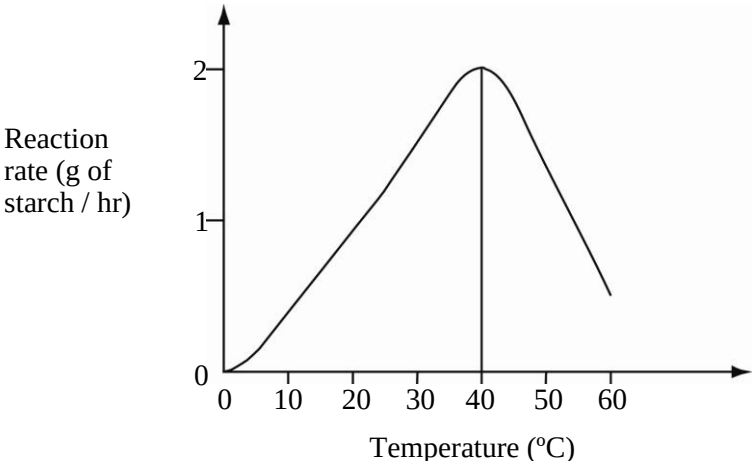
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	2
(c)	Understanding basic concepts	1
(d)(i)	Presenting information	2
(d)(ii)	Presenting information	2

|!|BCELC0050000023|!| **SI**

In an experiment, the effect of temperature on the breakdown of starch by amylase was investigated. The reaction rate is represented by the mass of starch broken down per hour. The following graph shows the results.

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(a) Describe the changes in the activity of the enzyme when the temperature increases from 0 ºC to 60 ºC? (3 marks)

(b) Why does the reaction rate drops rapidly at temperatures above 40 ºC? (1 mark)

(c) What is the product of the digestion of starch? (1 mark)

(d) How would you test whether this product is a reducing sugar? (2 marks)

(e) State ***one*** factor, other than temperature, that affects the rate of an enzymatic reaction. (1 mark)

##

(a) The enzyme activity increases from 0 ºC to 40 ºC. (1)
The optimum temperature of the enzyme is 40 ºC. (1)
The enzyme activity decreases from 40 ºC to 60 ºC. (1)

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(b) High temperatures cause denaturation and loss of functions in enzymes. (1)

(c) Maltose (1)

(d) Heat the reaction mixture with Benedict's solution. (1)
If a brick-red precipitate forms, it indicates that the product is a reducing sugar. (1)

(e) pH / substrate concentration / enzyme concentration / other reasonable answer (1)

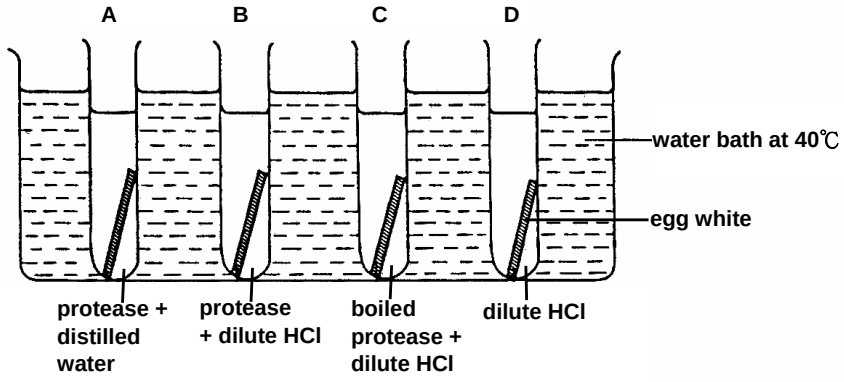
(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Practical skills	2
(e)	Understanding basic concepts	1

|!|BCELC0050000024|!| **SI** **STSE**

In an experiment investigating the action of a type of protease on protein, a bar of solid egg white was placed in each of the four test tubes of different contents. The test tubes were put into a water bath at 40 ºC, as shown in the following diagram.



The initial lengths of egg white were all 30 mm. After one hour, the length of egg white in each test tube was measured again. The following table shows the results.

Test tube	Length of egg white (mm)
A	29
B	25
C	30
D	30

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(a) Explain the results in tubes A, C and D. (3 marks)

(b) What conclusion can be drawn from the results of the experiment? (2 marks)

(c) Explain the following applications of protease in our daily lives.

(i) Biological washing powder can remove the egg stains on clothes. (2 marks)

(ii) When pineapple juice is added to a large piece of beef, the beef will become tenderer after 30 minutes. (2 marks)

##

(a) Tube A: Protease is not effective in a neutral medium. (1)
Tube C: Protease is denatured by boiling and becomes non-functional. (1)
Tube D : Dilute HCl has no digestive action on protein. (1)

(b) This kind of protease can digest protein in the presence of dilute HCl / in acidic condition (1)
but becomes ineffective after it is boiled. (1)

(c) (i) Protease in the biological washing powder digests protein in egg stain into simpler forms (1)
which are soluble in water and can be washed away by water. (1)

(ii) Beef is rich in fibrous protein. (1)
Protease in pineapple juice digests the fibrous protein in beef into simpler forms, making it tenderer. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Analysing data	2
(c)(i)	Applying concepts	2
(c)(ii)	Applying concepts	2

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|!|BCELB0050000025|!| ★ **STSE**

Wendy had prepared some fresh pineapples and was ready to use them to make a bowl of pineapple jelly for a party. However, the jelly solution failed to coagulate and remained in the liquid state even after it had cooled down.

She then searched for information on the Internet and found that jelly would become a semi-solid when the protein molecules inside formed bonded with each other.

(a) Which enzyme does fresh pineapple contain that prevents the jelly solution from solidifying? Explain how it worked. (3 marks)

(b) State ***two*** characteristics of the enzyme answered in (a). (2 marks)

(c) Wendy's classmate suggested that she should use canned pineapple instead of fresh pineapple, so that the jelly solution mixed with the pineapple can coagulate successfully. Comment on this method. (Hint: What treatment is applied in making canned pineapple?) (3 marks)

##

(a) Protease (1)
The protease from the fresh pineapple breaks down the bonds formed between the protein molecules in the jelly solution. (1)
In the absence of bonds that hold the protein molecules in a regular pattern, the jelly cannot turn from the liquid state into a semi-solid. (1)

(b) Any ***two*** of the following: (2)
- They are biological catalysts. (1)
- They are specific in action. (1)
- They are reusable. (1)

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(Or other reasonable answers)

(c) Pineapple is treated at a high temperature during the canning process. (1)
The proteases are denatured at the high temperature and (1)
they could not break the bonds formed between the protein molecules in the jelly solution. (1)
Therefore the jelly is able to solidify.

(8 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Understanding basic concepts	2
(c)	Applying concepts	3

|!|BCELB0050000026|!| ★ SI

The table below shows the contents in five test tubes in an experiment which demonstrates the digestive action of lipase.

	Test tubes				
Test tube contents	**A**	**B**	**C**	**D**	**E**
Whole fat milk (cm^3)	5	5	5	5	5
Sodium hydrogencarbonate solution (cm^3)	1	1	1	1	1
Bile salt solution (cm^3)	0	1	0	1	1
Lipase solution (cm^3)	0	0	1	1	1 (boiled)
Distilled water (cm^3)	2	1	1	0	0

After one hour, pH paper was used to test the pH of the solutions of the five test tubes. It was found that only solutions in test tubes C and D were acidic.

(a) Which substance present in the milk is the substrate of the reaction catalysed by lipase? (1 mark)

_____

(b) What is the purpose of adding sodium hydrogencarbonate solution to each test tube? (1 mark)

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(c) State the final product of the reaction which made the solution in test tubes C and D acidic. (1 mark)

(d) Compare and explain the acidity of the solutions in test tubes C and D. (3 marks)

(e) Explain the results of test tube E. (2 marks)

(f) Another test tube, F, contains a mixture of the following, was prepared.

Contents	Volume (cm^3)
Whole fat milk	5
Sodium hydrogencarbonate solution	1
Bile salt solution (boiled)	1
Lipase solution	1
Distilled water	0

The solution was found to be acidic after one hour.

(i) Which test tube can act as a control to test tube F? (1 mark)

(ii) Is bile salt an enzyme? Explain your answer. (2 marks)

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(a) Fat / lipid (1)

(b) To provide a slightly alkaline condition for the lipase to work. (1)

(c) Fatty acids (1)

(d) Solution in test tube D is more acidic than that in test tube C, because (1)
the bile salts in D emulsified the fat (broke the fat into smaller droplets) and
increased the surface area for the action of lipase. (1)

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With a larger surface area, the rate of breakdown reaction was faster in D and more fatty acids were formed. (1)

(e) The shape of the active site of the lipase was altered after boiling. (1)
The substrate could not fit into the active site to form the products and the lipid breakdown reaction is inhibited. (1)

(f) (i) Test tube D (1)

(ii) No, bile salt is not an enzyme. (1)
It is not denatured after boiling. (1)

(11 marks)

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	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Practical skills	1
(c)	Understanding basic concepts	1
(d)	Applying concepts	3
(e)	Applying concepts	2
(f)(i)	Analysing data	1
(f)(ii)	Applying concepts	2

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